Project Report: Home Expense Tracker

Track Name: CLI Project

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1. Introduction

The Home Expense Tracker is a command-line based application designed to help students or flatmates manage shared house expenses easily. It simplifies the process of calculating rent and dividing monthly bills among all members.

2. Features

1. **Object-Oriented Design**
   The project is built using classes such as -House, Room, ExpenseManager, CLIController
   This makes the system organized and easy to maintain.

2. **Dynamic Expense Input**
   The program asks the user to enter monthly shared expenses like: Wi-Fi, Bua, Waste, Electricity, Gas, Lift, Utility, Water.

3. **Per-Person Cost Calculation**
   The system automatically calculates the total number of people in the house and divides the total cost fairly among them.

4. **Clear Output Display**
   At the end, the program shows a simple and clean breakdown of:
   - Total expenses
   - Individual cost
   - Detailed payment information

3. Setup and Run Steps

1. **Check Requirements**

   Make sure Python is installed on your computer.

2. **Go to Project Folder**

   Open the terminal or command prompt and navigate to the project directory:

   Example- cd t:/6_Project/Home_Expense_Tracker/Python_Project/Home_Expense_Tracker/

3. **Run the Program**

   Execute the main file to start the program: python main.py

4. How the Program Works

- The program runs step by step like a guide.
- First, it asks for the number of rooms.
- Then for each room, it asks:
  -Room rent
  -Number of people staying
- After that, it asks for all shared monthly expenses.

5. Output

After entering all the data, the program generates a result that includes:

- Total monthly expense
- Total number of people
- Cost per person
- A clear breakdown of who needs to pay how much

6. **Demo**

Input:

```
PS C:\Users\acer> cd D:\Python\Python_Project\Home_Expense_Tracker
PS D:\Python\Python_Project\Home_Expense_Tracker> python main.py
Welcome!

Enter the number of room of your house?
3
Enter the rent of room 1
10000
Enter the number of persons in room 1
2
Enter the rent of room 2
12000
Enter the number of persons in room 2
2
Enter the rent of room 3
7000
Enter the number of persons in room 3
1
```

```
Enter the Wi-Fi bill:
1050
Enter the Bua bill:
4000
Enter the waste bill:
250
Enter the electricity bill:
1500
Enter the gas bill:
1500
Enter the lift bill:
1000
Enter the utility bill:
500
Enter the water bill:
1000
```

Output:

```
==========================================
             Final Cost Breakdown
==========================================

--- Shared House Bills ---
  Total combined bills:   10800.00
  Cost per person:        2160.00

--- Individual Room Breakdowns ---
  Room 1:
    Base Rent (per person): 5000.00
    Total Due (with bills): 7160.00

  Room 2:
    Base Rent (per person): 6000.00
    Total Due (with bills): 8160.00

  Room 3:
    Base Rent (per person): 7000.00
    Total Due (with bills): 9160.00

==========================================
PS D:\Python\Python_Project\Home_Expense_Tracker>
```

7. Conclusion

This project helps users easily manage shared living expenses without manual calculation. It reduces confusion, saves time, and ensures fair cost distribution among all house members.